

Sheet 1 Lab Report

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2024CSB041

1 Assignment 1

Q1: Constraint violation attempts

```
-- PRIMARY KEY violation
INSERT INTO
  depts (deptcode, deptname)
VALUES
  ('CSE', 'Duplicate Dept');

-- CHECK violation: bdate >= '1997-01-01'
INSERT INTO
  students (rollno, name, bdate, deptcode, hostel, parent_inc)
VALUES
  (
    99999999,
    'Test Student',
    '1998-05-10',
    'CSE',
    1,
    100000.0
  );

-- CHECK violation: hostel >= 10
INSERT INTO
  students (rollno, name, bdate, deptcode, hostel, parent_inc)
VALUES
  (
    99999998,
    'Test Student 2',
    '1995-01-01',
    'CSE',
    15,
    100000.0
  );

-- FOREIGN KEY violation: deptcode
INSERT INTO
  students (rollno, name, bdate, deptcode, hostel, parent_inc)
VALUES
  (
    99999997,
    'Test Student 3',
    '1995-01-01',
    'XYZ',
    1,
```

```
100000.0  
);
```

Error: duplicate key value violates unique constraint "depts_pkey" DETAIL: Key (deptcode)=(CSE) already exists.

Error: new row for relation "students" violates check constraint "students_bdate_check" DETAIL: Failing row contains (99999999, Test Student , 1998-05-10, CSE, 1, 100000.0).

Error: new row for relation "students" violates check constraint "students_hostel_check" DETAIL: Failing row contains (99999998, Test Student 2 , 1995-01-01, CSE, 15, 100000.0).

Error: insert or update on table "students" violates foreign key constraint "students_deptcode_fkey" DETAIL: Key (deptcode)=(XYZ) is not present in table "depts".

Q2: Delete deptcode='CSE' (cascades to students)

```
SELECT
    rollno,
    name
FROM
    students
WHERE
    TRIM(deptcode) = 'CSE';

DELETE FROM crs_regd
WHERE
    crs_cd IN (
        SELECT
            crs_code
        FROM
            crs_offrd
        WHERE
            crs_fac_cd IN (
                SELECT
                    fac_code
                FROM
                    faculty
                WHERE
                    TRIM(fac_dept) = 'CSE'
            )
    );

DELETE FROM crs_regd
WHERE
    crs_rollno IN (
        SELECT
            rollno
        FROM
            students
        WHERE
            TRIM(deptcode) = 'CSE'
    );

DELETE FROM crs_offrd
WHERE
    crs_fac_cd IN (
        SELECT
            fac_code
        FROM
            faculty
        WHERE
            TRIM(fac_dept) = 'CSE'
    );

DELETE FROM faculty
WHERE
    TRIM(fac_dept) = 'CSE';

DELETE FROM depts
WHERE
    TRIM(deptcode) = 'CSE';
```

```
SELECT
    rollno ,
    name
FROM
    students
WHERE
    TRIM(deptcode) = 'CSE';
```

Output: *CSE Students Before Deletion:*

rollno	name
92005001	Riddhi Agarwal
92005002	Priya Verma
92005003	Rahul Das
92005004	Sneha Gupta

Statement executed successfully.

Statement executed successfully.

Statement executed successfully.

Statement executed successfully.

Statement executed successfully.

Statement executed successfully. (0 rows returned)

Q3: Courses offered by faculty 'dbp' and 'nls'

```
SELECT
    crs_code ,
    crs_name
FROM
    crs_offrd
WHERE
    TRIM(crs_fac_cd) IN ('dbp', 'nls');
```

Output:

crs_code	crs_name
CS101	DBMS
CS102	OS
CS103	Software Engg.
CS104	MIS

Q4: Courses with full details offered by 'dbp'

```
SELECT
    co.crs_code ,
    co.crs_name ,
    co.crs_credits ,
    f.fac_name ,
    f.fac_dept
FROM
    crs_offrd co
    JOIN faculty f ON co.crs_fac_cd = f.fac_code
WHERE
    TRIM(co.crs_fac_cd) = 'dbp';
```

Output:

crs_code	crs_name	crs_credits	fac_name	fac_dept
CS103	Software Engg.	6.0	Dr. D.B. Phatak	CSE
CS102	OS	5.0	Dr. D.B. Phatak	CSE
CS101	DBMS	4.0	Dr. D.B. Phatak	CSE

Q5: Courses with credits between 4.0 and 6.0

```
SELECT
    crs_code ,
    crs_name ,
    crs_credits
FROM
    crs_offrd
WHERE
    crs_credits BETWEEN 4.0 AND 6.0;
```

Output:

crs_code	crs_name	crs_credits
CS101	DBMS	4.0
CS102	OS	5.0
CS103	Software Engg.	6.0
CS104	MIS	4.5
EE101	Circuit Theory	5.5
IT101	Data Structures	4.0

Q6: Courses with credits greater than 6.5

```
SELECT
    crs_code ,
    crs_name ,
    crs_credits
FROM
    crs_offrd
WHERE
    crs_credits > 6.5;
```

Output:

crs_code	crs_name	crs_credits
EE102	Signal Processing	7.0
PH106	Physics	8.0
ME101	Thermodynamics	7.5

2 Assignment 2

Q1: Count students in CSE dept

```
SELECT
  COUNT(*)
FROM
  students
WHERE
  TRIM(deptcode) = 'CSE';
```

Output:

count
4

Q2: Min, max and average marks of each course

```
SELECT
    crs_cd,
    MIN(marks) AS min_marks,
    MAX(marks) AS max_marks,
    AVG(marks) AS avg_marks
FROM
    crs_regd
GROUP BY
    crs_cd;
```

Output:

crs_cd	min_marks	max_marks	avg_marks
PH106	55.00	90.00	74.8750000000000000
CS104	91.00	91.00	91.0000000000000000
CH103	60.00	88.00	72.0000000000000000
CS101	30.00	92.00	56.0500000000000000
ME101	65.00	71.00	68.0000000000000000
EE101	70.00	82.00	75.2500000000000000
CS102	72.00	88.00	77.7500000000000000
IT101	76.00	82.00	79.0000000000000000
CS103	68.00	68.00	68.0000000000000000
EE102	60.00	85.00	72.5000000000000000

Q3: Total credits of courses registered by a student

```
SELECT
  r.crs_rollno ,
  s.name ,
  SUM(o.crs_credits) AS total_credits
FROM
  crs_regd r
  JOIN crs_offrd o ON r.crs_cd = o.crs_code
  JOIN students s on r.crs_rollno = s.rollno
GROUP BY
  r.crs_rollno ,
  s.name;
```

Output:

crs_rollno	name	total_credits
92005004	Sneha Gupta	9.0
92005012	Vikram Singh	7.0
92005103	Pooja Reddy	15.0
92005011	Neha Patil	9.5
92005001	Riddhi Agarwal	54.5
92005102	Deepak Mehta	16.5
92005003	Rahul Das	4.0
92005010	Ravi Kumar	9.5
92005100	Suresh Nair	20.0
92005002	Priya Verma	9.0
92005104	Arun Bhat	8.0
92005101	Kavita Jain	4.0
92005106	Kiran Desai	7.5

Q4: Number of students in each hostel whose department is CSE

```
SELECT
    hostel,
    COUNT(*) AS num_students
FROM
    students
WHERE
    TRIM(deptcode) = 'CSE'
GROUP BY
    hostel;
```

Output:

hostel	num_students
1	2
2	1
3	1

Q5: Hostel, rollno, parent_inc of student with max(parent_inc) in each hostel

```
SELECT
    hostel,
    rollno,
    parent_inc
FROM
    (
        SELECT
            hostel,
            rollno,
            parent_inc,
            RANK() OVER (
                PARTITION BY
                    hostel
                ORDER BY
                    parent_inc DESC
            ) as rank
        FROM
            students
        WHERE
            hostel IS NOT NULL
    ) ranked_students
WHERE
    rank = 1;
```

Output:

hostel	rollno	parent_inc
1	92005001	500000.0
2	92005002	750000.0
3	92005004	600000.0
4	92005011	900000.0
5	92005012	350000.0
6	92005100	800000.0
7	92005101	650000.0
8	92005103	700000.0
9	92005105	550000.0

Q6: Name and parent_inc of students with ncome greater than rollno 92005010

```
SELECT
    name ,
    parent_inc
FROM
    students
WHERE
    parent_inc > (
        SELECT
            parent_inc
        FROM
            students
        WHERE
            rollno = 92005010
    );
```

Output:

name	parent_inc
Riddhi Agarwal	500000.0
Priya Verma	750000.0
Sneha Gupta	600000.0
Neha Patil	900000.0
Ananya Roy	550000.0
Suresh Nair	800000.0
Kavita Jain	650000.0
Pooja Reddy	700000.0
Meera Iyer	550000.0
Kiran Desai	480000.0

Q7: Marks of students who scored more than rollno 92005102 in CH103 and PH106

```
SELECT
    r.crs_rollno,
    r.crs_cd,
    r.marks
FROM
    crs_regd r
WHERE
    TRIM(r.crs_cd) IN ('CH103', 'PH106')
    AND r.marks > (
        SELECT
            r2.marks
        FROM
            crs_regd r2
        WHERE
            r2.crs_rollno = 92005102
            AND r2.crs_cd = r.crs_cd
    );
```

Output:

crs_rollno	crs_cd	marks
92005001	CH103	88.00
92005001	PH106	74.50
92005012	CH103	65.00
92005100	CH103	75.00
92005100	PH106	80.00
92005103	PH106	90.00

3 Assignment 3

Q1: Students (rollno, name, deptcode) registered for course EE101

```
SELECT
    s.rollno,
    s.name,
    s.deptcode
FROM
    students s
JOIN crs_regd cr ON s.rollno = cr.crs_rollno
WHERE
    TRIM(cr.crs_cd) = 'EE101';
```

Output:

rollno	name	deptcode
92005001	Riddhi Agarwal	CSE
92005010	Ravi Kumar	ELE
92005011	Neha Patil	ELE
92005102	Deepak Mehta	ETC

Q2: Students (rollno, name) in ELE dept registered for course EE101

```
SELECT
  s.rollno,
  s.name
FROM
  students s
  JOIN crs_regd cr ON s.rollno = cr.crs_rollno
WHERE
  TRIM(s.deptcode) = 'ELE'
  AND TRIM(cr.crs_cd) = 'EE101';
```

Output:

rollno	name
92005010	Ravi Kumar
92005011	Neha Patil

Q3: Students (rollno, name) in ELE dept NOT registered for course EE101

```
SELECT
    rollno ,
    name
FROM
    students
WHERE
    TRIM(deptcode) = 'ELE'
    AND rollno NOT IN (
        SELECT
            crs_rollno
        FROM
            crs_regd
        WHERE
            TRIM(crs_cd) = 'EE101'
    );
```

Output:

rollno	name
92005012	Vikram Singh
92005013	Ananya Roy

Q4: Students registered for BOTH 'DBMS' and 'OS'

```
SELECT
    s.name
FROM
    students s
JOIN crs_regd cr1 ON s.rollno = cr1.crs_rollno
JOIN crs_offrd co1 ON cr1.crs_cd = co1.crs_code
JOIN crs_regd cr2 ON s.rollno = cr2.crs_rollno
JOIN crs_offrd co2 ON cr2.crs_cd = co2.crs_code
WHERE
    TRIM(co1.crs_name) = 'DBMS'
    AND TRIM(co2.crs_name) = 'OS';
```

Output:

name
Riddhi Agarwal
Priya Verma
Sneha Gupta
Suresh Nair

Q5: Faculty who offered either 'MIS' or 'Software Engg.'

```
SELECT
    f.fac_name
FROM
    faculty f
    JOIN crs_offrd co ON f.fac_code = co.crs_fac_cd
WHERE
    TRIM(co.crs_name) = 'MIS'
    OR TRIM(co.crs_name) = 'Software Engg.';
```

Output:

fac_name
Dr. D.B. Phatak
Dr. N.L. Sarda

Q6: Faculty who offered 'MIS' but NOT 'Software Engg.'

```
SELECT
    f.fac_name
FROM
    faculty f
JOIN crs_offrd co ON f.fac_code = co.crs_fac_cd
WHERE
    TRIM(co.crs_name) = 'MIS'
AND f.fac_code NOT IN (
    SELECT
        crs_fac_cd
    FROM
        crs_offrd
    WHERE
        TRIM(crs_name) = 'Software Engg.'
);
```

Output:

fac_name
Dr. N.L. Sarda

Q7: Students in each hostel not registered for any course

```
SELECT
  s.hostel,
  s.rollno,
  s.name
FROM
  students s
WHERE
  s.rollno NOT IN (
    SELECT
      crs_rollno
    FROM
      crs_regd
  );
```

Output:

hostel	rollno	name
4	92005013	Ananya Roy
9	92005105	Meera Iyer

Q8: Students in ELE dept OR registered for course CS101

```
SELECT
    rollno ,
    name
FROM
    students
WHERE
    TRIM(deptcode) = 'ELE'
UNION
SELECT
    s.rollno ,
    s.name
FROM
    students s
JOIN crs_regd cr ON s.rollno = cr.crs_rollno
WHERE
    TRIM(cr.crs_cd) = 'CS101';
```

Output:

rollno	name
92005001	Riddhi Agarwal
92005002	Priya Verma
92005003	Rahul Das
92005004	Sneha Gupta
92005010	Ravi Kumar
92005011	Neha Patil
92005012	Vikram Singh
92005013	Ananya Roy
92005100	Suresh Nair
92005101	Kavita Jain
92005104	Arun Bhat

Q9: Students registered for ALL courses

```
SELECT
    s.rollno,
    s.name
FROM
    students s
    JOIN crs_regd cr ON s.rollno = cr.crs_rollno
GROUP BY
    s.rollno,
    s.name
HAVING
    COUNT(DISTINCT cr.crs_cd) = (
        SELECT
            COUNT(*)
        FROM
            crs_offrd
    );
```

Output:

rollno	name
92005001	Riddhi Agarwal

Q10: Grace Marks +5 in 'DBMS' for students who scored ; 50

```
UPDATE crs_regd
SET
    marks = marks + 5
WHERE
    marks < 50
    AND crs_cd IN (
        SELECT
            crs_code
        FROM
            crs_offrd
        WHERE
            TRIM(crs_name) = 'DBMS'
    );
```

Statement executed successfully.